

Design and Development of Reusable Welding Jig for the VEGA Street Bike

This project focused on the design and development of a reusable welding jig to support the accurate assembly of the Vega Street Bike chassis during fabrication. The primary objective was to ensure precise alignment and positioning of chassis members while welding, thereby minimizing dimensional errors and improving repeatability in the manufacturing process. The jig structure was designed to provide sufficient rigidity and stability while allowing easy placement and removal of chassis components.

- Ensured precise chassis dimensional accuracy during assembly
- Reduced fabrication time and improved repeatability
- Reusable, modular jig design for multiple build cycles
- Full SolidWorks 3D CAD model with manufacturing drawings

Designing Phase

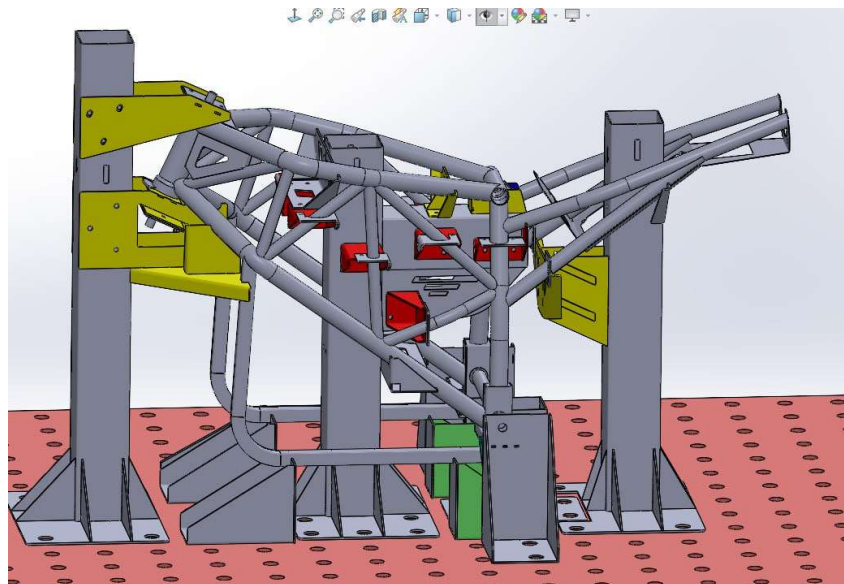


Figure 1-Full 3D CAD model of Jig with chassis

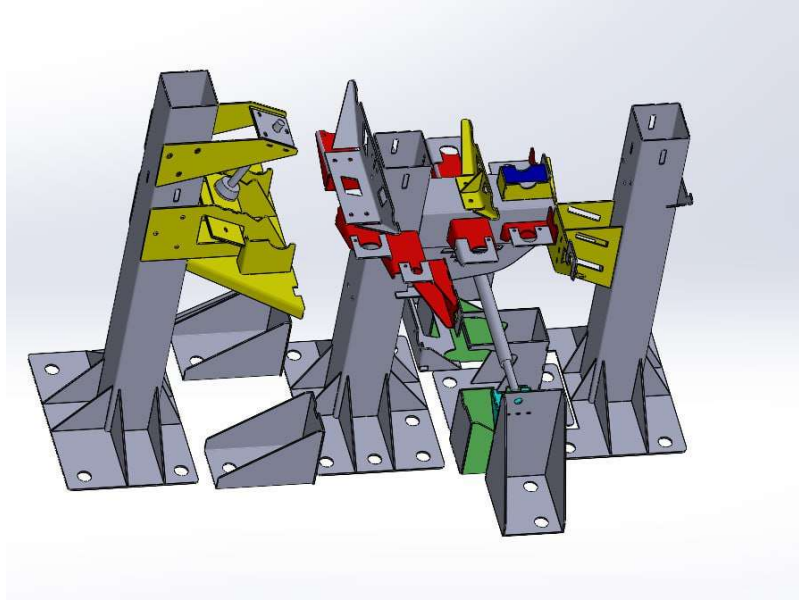


Figure 2 - 3D CAD model of the Jig

Manufacturing Phase

The welding jig was manufactured using sheet metal and box section bars, selected for their rigidity, weldability, and ease of fabrication. Sheet metal was utilized for base plates and mounting brackets, while box section bars formed the main structural skeleton of the jig. All components were cut, drilled, and welded in accordance with the SolidWorks manufacturing drawings, ensuring accurate positioning of all locating points and support surfaces.



Figure 3 - Sheet metal laser cutting



Figure 4 - Box bar cutting

Jig Assembly & Alignment Phase

The jig was precisely aligned using a laser level, ensuring accurate positioning across all axes before any chassis components were introduced. This guaranteed that all locating points and support surfaces were true to the intended design geometry prior to fabrication.



Figure 5 - Aligning using laser





Figure 6 - Jig Assembling



Figure 7 -Assembled Jig with chassis



Figure 8 -Assembled jig with chassis



Figure 9 -Fullu welded chassis